

The empirical recurrence for OEIS A252600, recovered from its tail

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8 September 2026

Abstract

OEIS A252600 records “Empirical recurrence of order 75” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 148$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 81$. Its last coefficient is $c_{75} = -442368 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A252600 is “Number of $(n+2) \times (3+2)$ 0..2 arrays with every 3×3 subblock row, column, diagonal and antidiagonal sum not equal to 1 2 or 5.”. It has offset 1 and begins

1218, 2784, 7769, 22623, 71730, 232461, 769153, 2611733, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:25:36, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 148$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 148$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 296$ exact terms from $n = 6$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	6	c_{20}	−1035806	c_{39}	−3572469431	c_{58}	−4376927510
c_2	−12	c_{21}	62652255	c_{40}	−5859484883	c_{59}	−3709761130
c_3	75	c_{22}	144865109	c_{41}	42916199	c_{60}	−982109320
c_4	−310	c_{23}	87977111	c_{42}	8738614547	c_{61}	1598124212
c_5	456	c_{24}	−173385238	c_{43}	11811914616	c_{62}	2287140704
c_6	−2155	c_{25}	−392845524	c_{44}	7155924312	c_{63}	1481678496
c_7	6227	c_{26}	−254784806	c_{45}	692351241	c_{64}	506702816
c_8	−5216	c_{27}	349666591	c_{46}	−1843271173	c_{65}	40148928
c_9	32890	c_{28}	442518485	c_{47}	−2831158819	c_{66}	−87962016
c_{10}	−56521	c_{29}	−375441956	c_{48}	−6784823138	c_{67}	−123783744
c_{11}	−6379	c_{30}	−1135721977	c_{49}	−10999769331	c_{68}	−118006848
c_{12}	−319140	c_{31}	160925079	c_{50}	−8766374060	c_{69}	−66567744
c_{13}	103392	c_{32}	3052549333	c_{51}	21604331	c_{70}	−17093376
c_{14}	555036	c_{33}	3471947228	c_{52}	7959941316	c_{71}	1090944
c_{15}	2339670	c_{34}	−200981778	c_{53}	8996005474	c_{72}	−610560
c_{16}	2682065	c_{35}	−4574532640	c_{54}	4896002558	c_{73}	−2916864
c_{17}	−3567366	c_{36}	−3961098877	c_{55}	923810051	c_{74}	−1769472
c_{18}	−14118868	c_{37}	−26863942	c_{56}	−1239375558	c_{75}	−442368
c_{19}	−28922112	c_{38}	631949202	c_{57}	−3036372986		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 81$, and it is the recurrence OEIS A252600 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{74} - \dots - c_{75}$. Its constant term is $-c_{75} = 442368$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -442368 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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